

PowerCube technology – scalable fast-charging system for e-trucks and e-buses

Battery-buffered ultra-fast charging system as a configurable modular platform: battery storage capacity, number of charging points and charging power are dimensioned to your requirements – from approx. 750 kWh and 400 kW in the compact configuration up to approx. 2 MWh of storage and 1 MW of charging power.

A SYSTEM, NOT A FIXED PRODUCT

Battery storage, power converters and charging technology in one enclosure – configured per project: capacity, power, number of charging points, charging interfaces (CCS/MCS).

TWO DEPLOYMENT SCENARIOS

Depot charging at the yard as well as public charging at rest stops or e-truck parking areas – stationary or mobile, grid-parallel or in island mode.

Agenda

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STARTING POINT

Two charging worlds – one bottleneck: the grid connection

DEPOT CHARGING – PREDICTABLE

- Long overnight dwell times; moderate power per charging point is sufficient.
- Many vehicles charging simultaneously create high load peaks at the site.
- The depot's grid connection is usually not sized for this – grid expansion is expensive and time-consuming.

PUBLIC CHARGING – TIME-CRITICAL

- Short dwell times, e.g. during the e-truck driver's mandatory break.
- High power per charging session, up to the megawatt range (MCS).
- Rest stops and parking areas often lack a sufficient grid connection.

The PowerCube decouples charging power from the grid connection: the storage unit charges continuously at limited grid power and delivers the energy to the vehicles at high power.

One system, three configuration levers

BATTERY STORAGE CAPACITY

up to ~2 MWh

Modular design based on battery strings — from approx. 750 kWh in the compact configuration up to approx. 2,000 kWh.

CHARGING POWER

up to 1 MW

From the 400 kW compact system to the megawatt class — depending on the converter and charging technology configuration.

CHARGING POINTS

2 to 8

Two integrated CCS charging points in the 10-foot variant and up to 8 external charging points via MCS or CCS dispensers with the MW charging variant.

Not a fixed product: each configuration is engineered to requirements — including grid connection (CEE 5/125 A or direct grid connection), operating mode (grid-parallel or island) and installation (stationary or mobile).

From the compact system to the megawatt class

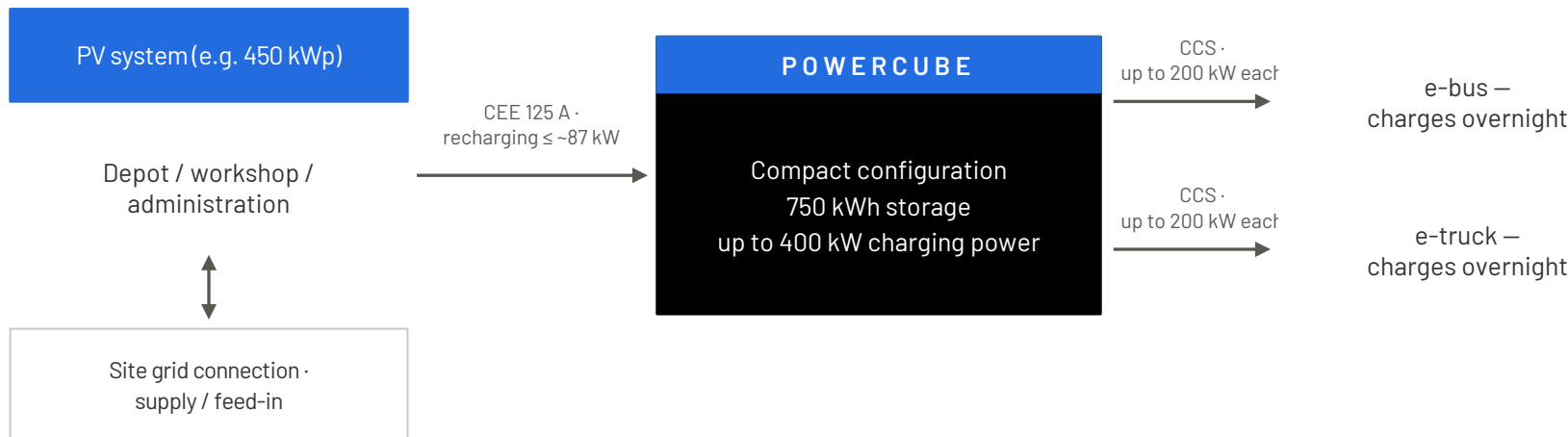
Parameter	PowerCube with Alpitronic HYC 400 – e.g. depot	MW PowerCube with Alpitronic HYC1000 – e.g. rest stop
Battery storage (gross)	approx. 750 kWh (LFP, 4 strings)	up to approx. 2,000 kWh (8 strings of approx. 250 kWh each)
Max. charging power	400 kW	up to 1,000 kW
Charging points	2× CCS, up to 200 kW each simultaneously	MCS + EV dispensers, up to 4 dispensers in total
Charging interfaces	CCS2, CHAdeMO; optionally CCS1, NACS, GB/T	MCS (Megawatt Charging System) + CCS
Grid connection	CEE 5/125 A or direct grid connection	Direct grid connection
Operating modes	Grid-parallel with peak shaving or island mode	Grid-parallel or island mode
Installation	Stationary or mobile	Stationary
Dimensions	2,990 × 2,500 × 2,760 mm, approx. 9.1 t	9,315 × 2,500 × 2,751 mm

Example configurations – intermediate levels of storage capacity, number of charging points and power are engineered per project.

DEPOT DEPLOYMENT

Depot charging: e-buses and e-trucks overnight

Solar power supplies the depot and additionally charges the PowerCube via a CEE socket. The vehicles charge at up to 2×200 kW directly from the storage unit – without upgrading the grid connection.



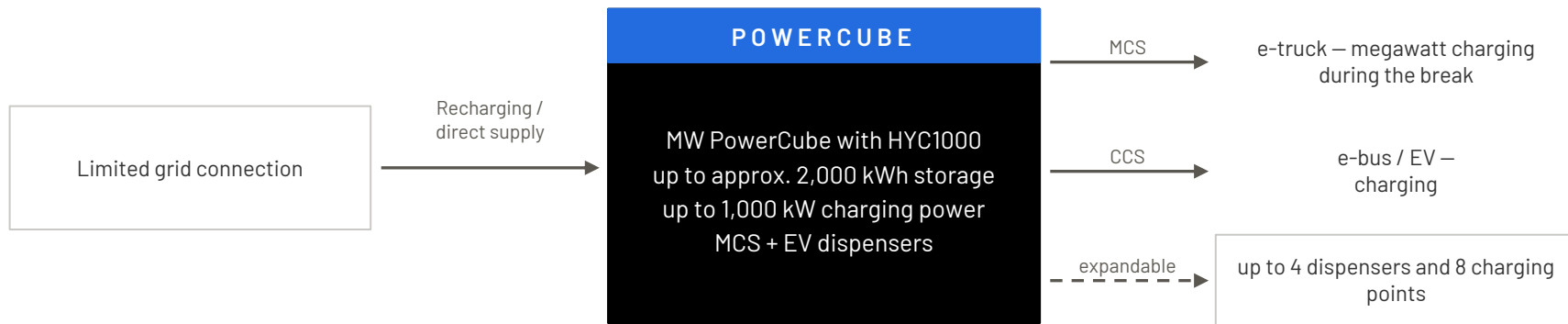
Power allocation selectable as “First Come First Serve” or “Equal Share”; the peak-shaving limit is configured in the web app and is never exceeded.

Schematic diagram – simplified, without protection and metering equipment.

PUBLIC DEPLOYMENT

Rest stop and e-truck parking: charging during the driving break

For public sites the system is configured larger: the MW PowerCube combines up to approx. 2 MWh of storage and up to 1 MW of charging power – with an MCS dispenser for e-trucks and EV dispensers for e-buses and other EVs.



Grid expansion with a high-voltage connection for conventional charging technology: planning and permitting can take up to 10 years.

Schematic diagram — simplified, without protection and metering equipment.

Source: HYC1000 PROMO operating and maintenance manual, AW-automotive GmbH (June 2025) · nationale-leitstelle.de/aktuelles-service/newshub/einfach-laden-an-rastanlagen-neue-studie-zur-auslegung-von-netzanschluesen-fuer-e-lkw-lade-hubs-veroeffentlicht/

STORAGECUBE – PURE BATTERY STORAGE

Storage today, charging infrastructure later

For companies and freight carriers that have not yet committed to e-trucks: the StorageCube is the storage system without integrated charging infrastructure – built from the same modules as the PowerCube (modular platform).

BENEFIT TODAY

- Energy management system for supporting the building grid and controlling connected loads.
- Grid-parallel operation on the low-voltage grid (3 AC 400 V) or self-sufficient island mode – backup power supplied entirely from the storage unit.
- Can be deployed stationary or mobile; installation without civil works.

EXPAND TOMORROW

- Once the e-truck decision is made, the site is expanded with PowerCubes including charging technology – the investment is preserved.
- Proven in combination: Audi charging hub with 3 PowerCubes, 1 StorageCube and 1 ControlCube on just a 200 kW grid connection.
- Outlook: bidirectional charging and grid services (frequency containment reserve, virtual power plant).

Series product since 2021: all components and the complete system certified – energy management system included.

TECHNICAL DATA

Battery system & charging – scalable

BATTERY SYSTEM

Energy capacity	approx. 750 kWh (LFP, 4 strings) up to approx. 2,000 kWh (8 strings of approx. 250 kWh each)
Voltage range	approx. 680 – 800 V DC (SOC-dependent)
Thermal management	TMS units keep the modules within the optimum temperature window
Safety	BMS monitors 12 parameters with early-warning levels; emergency stop, interlock, key switch per string
Redundancy	If one string shuts down, operation continues via the remaining strings – with derating if necessary
Battery warranty	SOH > 70% after 3,000 full cycles (compact configuration)

CHARGING TECHNOLOGY

HYC400 – COMPACT CONFIGURATION

Up to 400 kW (2× 200 kW simultaneously), 150–1,000 V DC; CCS2, CHAdeMO – optionally CCS1, NACS, GB/T; efficiency up to 97.5%.

HYC1000 – MW configuration

Hyperunit with up to 1,000 kW; MCS dispenser for e-trucks, CCS dispensers e.g. for e-buses; expandable to up to four dispensers = 8 charging points

HOW IT WORKS

How the PowerCube technology works

1

Grid charges the storage

The battery storage is continuously recharged via the grid connection (CEE 5/125 A or direct grid connection) – even in parallel with charging operations.

2

Storage delivers the peak power

During vehicle charging, battery and grid jointly feed the charging technology: up to 1 MW, far above the grid connection capacity.

3

EMS distributes the power

The energy management system always supplies the charging points with the maximum available power – automatic derating applies if it is exceeded.

4

Grid as fallback

If the storage is depleted, the charging points are automatically limited to the configured maximum grid connection power.

OPERATING MODES

Grid-parallel with peak shaving

Type and maximum power of the grid connection are configured; the set limit is never exceeded.

Island mode

Operation without a supply grid: all loads are powered exclusively from the storage unit.

Power allocation

Selectable “First Come First Serve” or “Equal Share” between the charging points.

Protection functions & certifications

BATTERY MANAGEMENT

Continuous monitoring of cell voltage, current and temperature with early-warning levels; automatic string shutdown in the event of a fault, interlock testing of all HV plug connections and immediate emergency-stop shutdown.

FIRE PROTECTION

Visual and acoustic alarm at critical thresholds, fire alarm control panel with declaration of conformity, plus water-spray extinguishing and fire alarm system in accordance with VdS 2395-1, EN 54-2 and EN 54-4.

CERTIFICATIONS

TÜV testing per VDE-AR 2510-50; grid integration VDE-AR-N 4105/4110; transport UN 38.3; BMS E1 (TÜV + KBA, ECE-R10); battery IEC 62619, UL 1973, UL 9540A; charging station IEC 61851, TÜV Süd CB; CE conformity.

Redundancy in operation: If one battery string shuts down, the remaining strings continue to supply the loads – with automatically reduced charging power if necessary.

CONCLUSION

One system for depot and rest stop

As a modular platform, the PowerCube scales from approx. 750 kWh and 400 kW up to approx. 2 MWh and 1 MW – charging e-trucks and e-buses where the grid ends: at the depot overnight and in public spaces during the driving break. Operators not yet running e-trucks start with the StorageCube as pure battery storage – and retrofit the charging technology later.

DELIVERY

Delivery time approx. 16 weeks from order.

WARRANTY

24 months from delivery.

BATTERY WARRANTY

SOH > 70% after 3,000 full cycles (DoD 90%, $P < 0.5 P$, < 2 cycles/day).

PROVIDER & CONTACT

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Concepts for battery-storage-backed charging infrastructure



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